

A purely odd representation of negative superdimension

Disproof of Conjecture 00000007400

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Claim and conventions

The original conjecture asserts nonnegativity of superdimensions of supersymmetric representations. Neither the English nor Chinese statement restricts the grading, excludes trivial actions, or imposes a positivity condition on the character. For a finite-dimensional complex super vector space $V = V_{\bar{0}} \oplus V_{\bar{1}}$, its standard superdimension is

$$\text{sdim } V = \dim_{\mathbb{C}} V_{\bar{0}} - \dim_{\mathbb{C}} V_{\bar{1}}.$$

Unlike ordinary dimension, this difference can be negative.

A complete superrepresentation

Let $\mathfrak{g}$ be the one-dimensional abelian complex Lie algebra viewed as a Lie superalgebra concentrated in even degree: $\mathfrak{g}_{\bar{0}} = \mathbb{C}$, $\mathfrak{g}_{\bar{1}} = 0$, and bracket zero. Bilinearity, graded skew symmetry and the graded Jacobi identity all hold because every bracket vanishes.

Let the underlying vector space of V be $\mathbb{C}$, with grading $V_{\bar{0}} = \{0\}$ and $V_{\bar{1}} = \mathbb{C}$. These are actual complex subspaces, disjoint and summing to V . Define the action $\rho(a)v = 0$ for every $a \in \mathfrak{g}$ and $v \in V$. It is bilinear and preserves parity. Its representation identity is

$$\rho([a, b])v = \rho(a)\rho(b)v - \rho(b)\rho(a)v = 0.$$

Since all nonzero algebra elements are even, this is exactly the required superrepresentation identity; requirements involving odd algebra elements hold trivially. Thus V is a finite-dimensional graded representation, not merely an assigned pair of dimension counts.

Its dimensions are $\dim V_{\bar{0}} = 0$ and $\dim V_{\bar{1}} = 1$, so

$$\text{sdim } V = 0 - 1 = -1 < 0.$$

This disproves the universal nonnegativity assertion. The same example is the parity reversal of the one-dimensional trivial even representation. Its supercharacter at the identity is the supertrace of the identity, equal to -1 ; character terminology does not turn it into a positive number.

Formal verification

The Lean project models the actual complex subspaces and their direct decomposition, the zero bilinear Lie bracket and its Jacobi identity, and bilinear actions with the commutator identity and parity preservation. The purely odd line is constructed as a structure satisfying these laws. Its dimensions are computed using Mathlib's vector-space `Module.finrank`, not introduced as hypotheses. The integer difference is proved equal to -1 . The final theorem `conjecture_7400_false` refutes nonnegativity even in this restricted class of superrepresentations, which is contained in the original unrestricted domain.